

Technical Challenge

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1 Method

1.1 Population and missing value imputation

The study cohort comprised 203 patients diagnosed with pancreatic ductal adenocarcinoma (PDAC), including 103 females and 100 males. To address missing data, we set a threshold of 10%, whereby variables with less than 10% missing values were considered suitable for imputation. Two variables met this criterion: neo-adjuvant therapy and type of surgery. The neo-adjuvant therapy variable is binary (yes/no). We employed logistic regression for imputation, utilizing other relevant variables in the dataset as predictors to estimate the missing values. This method leverages the relationships between variables to provide more accurate imputations compared to simpler methods like mean or mode imputation. The type of surgery variable is non-ordinal categorical. We applied polynomial regression for imputation, treating the categorical outcomes appropriately to predict the missing values. This approach allows for the modeling of complex relationships between variables, enhancing the accuracy of the imputations.

1.2 Normality assessment

To evaluate the distribution of 'total muscle volume' and 'total adipose tissue volume', we employed both graphical and statistical methods. Histograms were generated to visualize the frequency distributions, and Quantile-Quantile (Q-Q) plots were utilized to compare the observed data quantiles against the expected quantiles under a normal distribution. Following these visual assessments, the Shapiro-Wilk test was conducted to statistically assess the normality of these variables. This combination of graphical and statistical approaches provides a comprehensive evaluation of data normality.

1.3 Assessment of clinical therapy

To explore the relationships between the receipt of 'neoadjuvant therapy' (versus no neoadjuvant therapy) and 'adjuvant therapy' (versus no adjuvant therapy) with various body composition parameter, we conducted multiple linear regression analyses. The dependent variables in these analyses included 'total muscle

volume', 'total adipose tissue volume', 'intramuscular adipose tissue volume', 'subcutaneous adipose tissue volume', and 'visceral adipose tissue volume'. This analysis aimed to determine whether the administration of neoadjuvant or adjuvant therapy is associated with changes in body composition parameters.

1.4 Postoperative complication determination

For each patient, the ratio of intramuscular adipose tissue volume to total muscle volume was calculated. A ROC curve was created to detect if a specific cutoff value for this ratio is associated with increased postoperative complication rates. To evaluate the relationship between the ratio and postoperative complications, we performed logistic regression analysis. Gender was included as a covariate to adjust for potential confounding effects. This analysis aimed to determine whether a higher ratio is an independent predictor of postoperative complications.

1.5 Survival Analysis

To identify which body composition parameters have the most significant impact on overall survival, we constructed a multivariable Cox proportional hazards regression model. Variables considered in the model included total adipose tissue volume, bone density, and other relevant body composition metrics. Model selection was guided by the Akaike Information Criterion (AIC), with the goal of identifying the model that best balances goodness-of-fit and complexity. The AIC is calculated as follows: $AIC = -2 \times (\text{partial log-likelihood}) + 2 \times k$, where k represents the number of model parameters.

By implementing these statistical methods, we aimed to comprehensively assess the normality of key variables, evaluate the impact of therapeutic interventions on patient survival, and explore potential associations between therapies and body composition metrics.